Recap: Lecture 26

In Lecture 26, we

- Introduced the 1D wave equation in the context of modelling vibrations of a guitar string
- Discussed the fundamental differences between the wave equation and the heat equation, in terms of the physical implications and the genuine nature of the SoV solutions

Lecture 27

In Lecture 27, we will:

1. Introduce normal modes of the wave equation
2. Model vibrations of a drum lid using the 2D wave equation and apply SoV

Normal modes

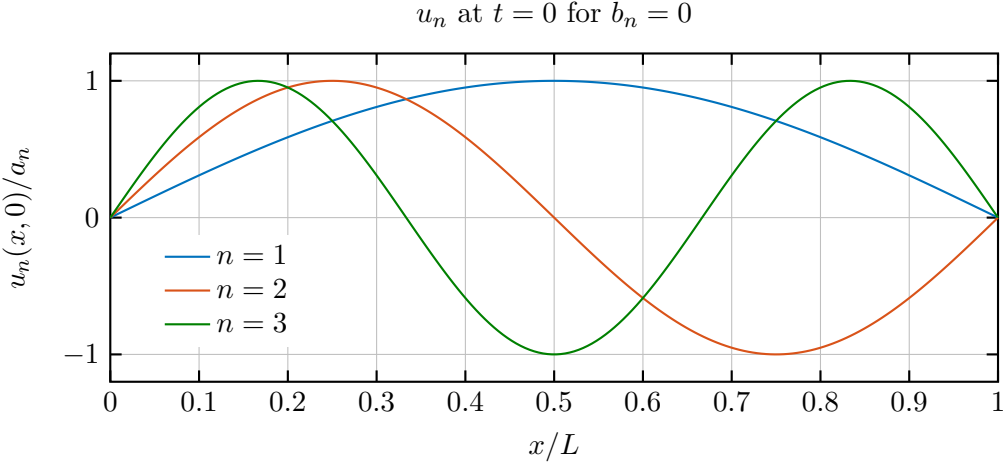
- At Step 4 of SoV for the guitar string (before applying ICs), we had

$$u(x,t) = \sum_{n=1}^{\infty} \underbrace{\{a_n \cos(\omega_n t) + b_n \sin(\omega_n t)\} \sin(k_n x)}_{u_n(x,t) = X_n(x)\, T_n(t)}$$

- $u_n(x,t)$ is a *normal mode* of the problem (1D wave eqn + Dirichlet BCs)
- Can refer to $\omega_n = ck_n$ as *normal frequencies* (or resonant frequencies)
- Can superpose any finite number of normal modes to construct a genuine solution

Normal modes

$$u_n(x,t) = \left\{a_n \cos(\omega_n t) + b_n \sin(\omega_n t)\right\} \sin(k_n x)$$



Wave equation: Problem 2

Find the normal modes of a rectangular drum lid

Governing PDE: 2D wave equation (membrane eqn) in Cartesian coordinates (x, y) is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad \text{for} \quad 0 < x < L_x, \;\; 0 < y < L_y, \;\; t > 0$$

Boundary conditions: Dirichlet BCs

$$u(x, 0, t) = u(x, b, t) = 0 \quad \text{for} \quad 0 < x < L_x, \; t > 0$$

$$u(0, y, t) = u(a, y, t) = 0 \quad \text{for} \quad 0 < y < L_y, \; t > 0$$

Initial conditions: Not required

Solving the 2D wave equation using separation of variables

1. Separate variables to turn single PDE into multiple ODEs

Use $u(x,t) = X(x)\,Y(y)\,T(t)$ in the 2D wave equation to derive the three ODEs

$$X'' - \mu\,X = 0\,, \quad Y'' - \nu\,Y = 0 \quad \text{and} \quad \ddot{T} - c^2(\mu+\nu)\,T = 0,$$

where λ and μ are two separation constants.

2. Apply the boundary conditions to find the eigenvalues and eigenfunctions.

The eigenvalue problems for X and Y are identical in form to those tackled for the 1D heat equation Problem 1.

Therefore, we have eigenfunctions and corresponding eigenvalues

$$X_n(x) = \sin(k_{x,m}x)\ , \quad k_{x,m} = \frac{\pi}{L_x}m\ , \quad \mu_m = -k_{x,m}^2 \quad (m = 1, 2, \ldots)$$

and

$$Y_n(y) = \sin(k_{y,n}y), \quad k_{y,n} = \frac{\pi}{L_y}n\ , \quad \nu_n = -k_{y,n}^2 \quad (n = 1, 2, \ldots)$$

3. Solve the remaining ODEs for each eigenvalue.

Let

$$\ddot{T}_{m,n} + \omega_{m,n}^2 T_{m,n} = 0 \quad \Rightarrow \quad T_{m,n}(t) = a_{m,n} \cos(\omega_{m,n} t) + b_{m,n} \sin(\omega_{m,n} t)$$

for $m, n = 1, 2, \ldots$, where

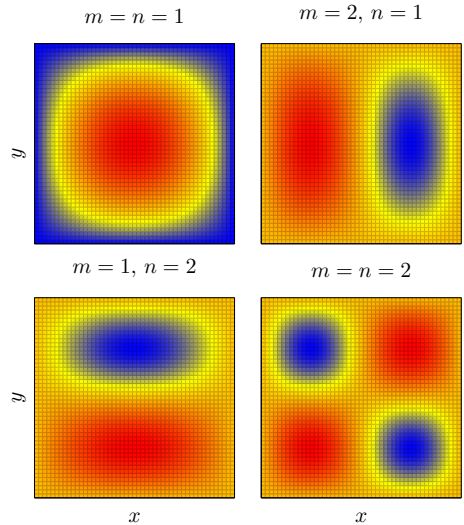
$$\omega_{m,n}^2 = -c^2(\mu_m + \nu_n) = c^2(k_{x,m}^2 + k_{y,n}^2)$$

This equation relating wave numbers and frequencies is called a dispersion relation.

Normal modes:

$$
\begin{aligned}
u_{m,n}(x,y,t) &= T_{m,n}(t)\, X_m(x)\, Y_n(y) \\
&= \{a_{m,n}\, \cos(\omega_{m,n} t) + \\
&\qquad b_{m,n}\, \sin(\omega_{m,n} t)\} \times \\
&\qquad \sin(k_{x,m} x) \times \\
&\qquad \sin(k_{y,n} y)
\end{aligned}
$$

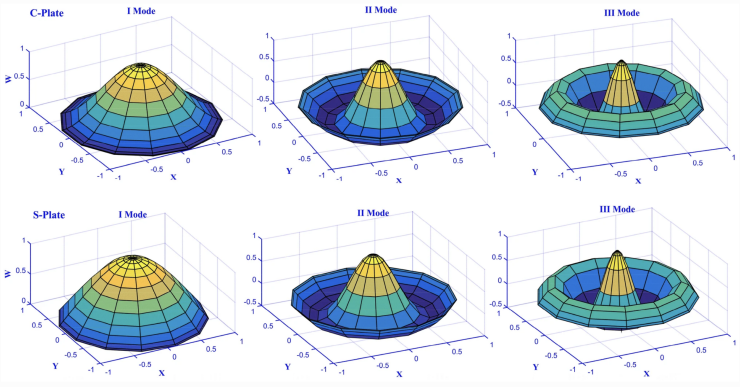
The figure assumes $L_x = L_y = 1$



Axisymmetric vibrations

- Axisymmetric normal modes of a circular drum lid using $J_0(r)$
  [https://en.wikipedia.org/wiki/Vibration_of_a_circular_membrane]

From: Axisymmetric vibrations of temperature-dependent functionally graded moderately thick circular plates with two-dimensional material and temperature distribution



Three-dimensional mode shapes for C and S-plates

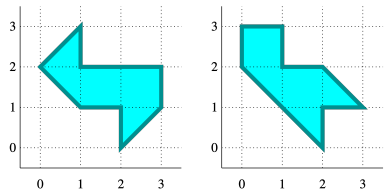
From: Saini & Lal, 2022. Engineering with Computers doi.org/10.1007/s00366-020-01056-1

Can one hear the shape of a drum?

In an influential article in the American Mathematical Monthly, Mark Kac asked in 1966 whether the normal frequencies determine the shape of the bounding region.
[https://doi.org/10.2307/2313748]

The problem was only solved (in the negative) in 1992 by Carolyn Gordon, David Webb, and Scott Wolpert [https://arxiv.org/abs/math/9207215]

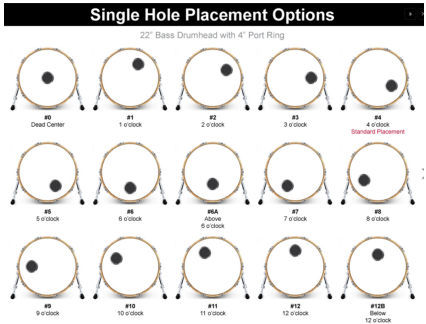
Isospectral drums:

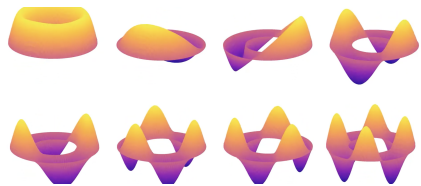


Under quite general assumptions, knowledge of the (asymptotics) of eigenvalues still permits to find the area and the circumference of the drum ("Weyl's formula").
[https://en.wikipedia.org/wiki/Hearing_the_shape_of_a_drum]

A hole in the drum

- Normal modes of drum lids with holes (not from SoV)



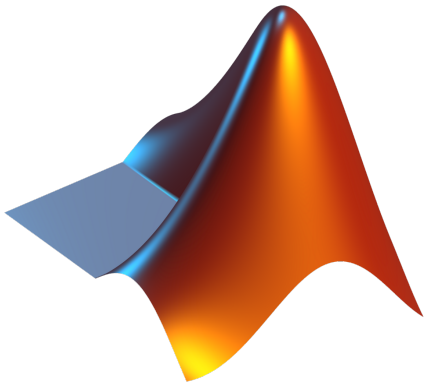


From https://www.vintagelogos.com/design-guide/bass-drum-port-holes/
https://www.youtube.com/watch?v=VjLaOTd1V08&list=PLt9eabBR9rbDASGaRJw5U2WdsctzuAvgE&index=8

MathWorks/MATLAB logo

The MathWorks logo is a lighted, reflective, surface plot of a variant of an eigenfunction of the L-shaped membrane. While the logo itself is a familiar icon, its mathematical background is less well known.

When a picnic basket covers one quarter of a square beach blanket, the remaining three quarters of the blanket form an L-shaped membrane. On a windy day, the motion of this membrane can be described using a wave equation.



https://au.mathworks.com/company/technical-articles/the-l-shaped-membrane.html